

Gabriel Castillo

04/26/2024

CS470 Final Reflection

This course gives valuable insight into how containers are used to run multiple applications separate from each other. The same OS and kernel are used efficiently. Aside from containers, this course shows how to fork an application via command prompt/power shell, then run the application. Afterwards, migrate the full stack application to a serverless cloud-native platform like AWS. And all the microservices needed to run the application ranging from Lambdas, S3, API Gateway, and DynamoDB. These skills will assist me in developing applications within similar serverless platforms.

My strength as a software engineer is finding better ways to keep code concise. For example, if there is a possibility of 30 lines of code to compress to 20, then I will figure out some way. I feel comfortable writing algorithms, sequence diagrams, and programming (this includes software automation). Throughout the whole Computer Science program, especially the programming assignments, a strategy was necessary to efficiently use my time to create a program. At times, I would begin with algorithms, but I still used sequence diagrams and flowcharts to simplify the problem.

Serverless scales automatically based on incoming request. The cloud also provides the infrastructure for error handling. You may need to create roles, policies, and permissions. Cost patterns will reflect the actual usage of the service, so the application will only use as much as is needed. Now this is considering if there is a sudden spike in traffic, the comparison is still efficient.

The expansion will include elasticity, this will automatically scale for the application which diminishes downtime. The global reach is far, as databases are scattered across the world, therefore, reaching a larger audience. And the negative of global outreach is data security and compliance. Many companies need to stay up to date with laws and new threats. Another con is relying too much on a specific cloud provider. When a migration is needed to further expand, it makes the task harder.

Elasticity provides flexibility and efficiency in data use. Pay-per-service is based on possible usage and traffic patterns. If an application has planned growth, then, this service would be best.

Youtube Video link:

<https://youtu.be/cp1PEvJfcl8>